

# Complete `firewall-cmd` Command Examples Guide

## Introduction

`firewall-cmd` is the command-line interface for managing `firewalld` on Linux systems such as:

- Rocky Linux
  - CentOS
  - RHEL
  - Fedora
  - AlmaLinux
- 

## 1. Basic Firewalld Service Management

### Check firewalld status

```
systemctl status firewalld
```

### Start firewalld

```
sudo systemctl start firewalld
```

### Stop firewalld

```
sudo systemctl stop firewalld
```

### Restart firewalld

```
sudo systemctl restart firewalld
```

### Enable firewalld at boot

```
sudo systemctl enable firewalld
```

## Disable firewalld at boot

```
sudo systemctl disable firewalld
```

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## 2. Basic Firewall Information

### Check firewall state

```
sudo firewall-cmd --state
```

### Reload firewall rules

```
sudo firewall-cmd --reload
```

### Reload completely

```
sudo firewall-cmd --complete-reload
```

### List all active zones

```
sudo firewall-cmd --get-active-zones
```

### List all available zones

```
sudo firewall-cmd --get-zones
```

### Display default zone

```
sudo firewall-cmd --get-default-zone
```

## Set default zone

```
sudo firewall-cmd --set-default-zone=public
```

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## 3. Zone Management

### List all settings in current zone

```
sudo firewall-cmd --list-all
```

### List all settings in specific zone

```
sudo firewall-cmd --zone=public --list-all
```

### Create a new zone

```
sudo firewall-cmd --permanent --new-zone=internal
```

### Delete a zone

```
sudo firewall-cmd --permanent --delete-zone=internal
```

### Reload after changes

```
sudo firewall-cmd --reload
```

### Assign interface to zone

```
sudo firewall-cmd --zone=internal --change-interface=eth1
```

## Permanently assign interface to zone

```
sudo firewall-cmd --permanent --zone=internal --add-interface=eth1
```

## Remove interface from zone

```
sudo firewall-cmd --permanent --zone=internal --remove-interface=eth1
```

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# 4. Service Management

## List allowed services

```
sudo firewall-cmd --list-services
```

## Add HTTP service temporarily

```
sudo firewall-cmd --add-service=http
```

## Add HTTPS service permanently

```
sudo firewall-cmd --permanent --add-service=https
```

## Remove HTTP service

```
sudo firewall-cmd --remove-service=http
```

## Remove HTTPS service permanently

```
sudo firewall-cmd --permanent --remove-service=https
```

## Add SSH service

```
sudo firewall-cmd --permanent --add-service=ssh
```

## Add FTP service

```
sudo firewall-cmd --permanent --add-service=ftp
```

## Add DNS service

```
sudo firewall-cmd --permanent --add-service=dns
```

## Add Samba service

```
sudo firewall-cmd --permanent --add-service=samba
```

## Reload firewall

```
sudo firewall-cmd --reload
```

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# 5. Port Management

## Open TCP port temporarily

```
sudo firewall-cmd --add-port=8080/tcp
```

## Open UDP port temporarily

```
sudo firewall-cmd --add-port=53/udp
```

## Open TCP port permanently

```
sudo firewall-cmd --permanent --add-port=8080/tcp
```

## Open port range

```
sudo firewall-cmd --permanent --add-port=6000-6010/tcp
```

## Remove open port

```
sudo firewall-cmd --permanent --remove-port=8080/tcp
```

## List open ports

```
sudo firewall-cmd --list-ports
```

## Check if port is open

```
sudo firewall-cmd --query-port=8080/tcp
```

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# 6. Rich Rules Examples

## Allow specific IP address

```
sudo firewall-cmd --permanent  
--add-rich-rule='rule family="ipv4" source address="192.168.1.100" accept'
```

## Block specific IP address

```
sudo firewall-cmd --permanent  
--add-rich-rule='rule family="ipv4" source address="10.10.10.50" reject'
```

## Allow SSH from specific subnet

```
sudo firewall-cmd --permanent  
--add-rich-rule='rule family="ipv4" source address="192.168.1.0/24" service  
name="ssh" accept'
```

## Allow port from specific IP

```
sudo firewall-cmd --permanent  
--add-rich-rule='rule family="ipv4" source address="192.168.1.20" port  
protocol="tcp" port="3306" accept'
```

## Drop all traffic from IP

```
sudo firewall-cmd --permanent  
--add-rich-rule='rule family="ipv4" source address="172.16.1.100" drop'
```

## Limit SSH connections

```
sudo firewall-cmd --permanent  
--add-rich-rule='rule service name="ssh" limit value="5/m" accept'
```

## Log and drop packets

```
sudo firewall-cmd --permanent  
--add-rich-rule='rule family="ipv4" source address="192.168.50.5" log  
prefix="BlockedIP" level="info" drop'
```

## View rich rules

```
sudo firewall-cmd --list-rich-rules
```

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## 7. Source IP Management

### Add source IP to zone

```
sudo firewall-cmd --permanent --zone=trusted --add-source=192.168.1.0/24
```

### Remove source IP

```
sudo firewall-cmd --permanent --zone=trusted --remove-source=192.168.1.0/24
```

### List sources

```
sudo firewall-cmd --zone=trusted --list-sources
```

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## 8. Masquerading and NAT

### Enable masquerading

```
sudo firewall-cmd --permanent --add-masquerade
```

### Disable masquerading

```
sudo firewall-cmd --permanent --remove-masquerade
```

### Check masquerading status

```
sudo firewall-cmd --query-masquerade
```

### Port forwarding example

```
sudo firewall-cmd --permanent  
--add-forward-port=port=8080:proto=tcp:toport=80
```



## Forward to another IP

```
sudo firewall-cmd --permanent  
--add-forward-port=port=2222:proto=tcp:toport=22:toaddr=192.168.1.10
```

## Remove forwarding rule

```
sudo firewall-cmd --permanent  
--remove-forward-port=port=8080:proto=tcp:toport=80
```

---

# 9. ICMP Management

## List ICMP types

```
sudo firewall-cmd --get-icmptypes
```

## Block ping requests

```
sudo firewall-cmd --permanent --add-icmp-block=echo-request
```

## Remove ICMP block

```
sudo firewall-cmd --permanent --remove-icmp-block=echo-request
```

## List ICMP blocks

```
sudo firewall-cmd --list-icmp-blocks
```

---

## 10. Panic Mode

### Enable panic mode

```
sudo firewall-cmd --panic-on
```

### Disable panic mode

```
sudo firewall-cmd --panic-off
```

### Query panic mode status

```
sudo firewall-cmd --query-panic
```

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## 11. Runtime vs Permanent Rules

### Runtime rule example

```
sudo firewall-cmd --add-service=http
```

### Permanent rule example

```
sudo firewall-cmd --permanent --add-service=http
```

### Copy runtime configuration to permanent

```
sudo firewall-cmd --runtime-to-permanent
```

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## 12. Direct Rules

### Add direct iptables rule

```
sudo firewall-cmd --direct --add-rule ipv4 filter INPUT 0  
-p tcp --dport 1234 -j ACCEPT
```

### Remove direct rule

```
sudo firewall-cmd --direct --remove-rule ipv4 filter INPUT 0  
-p tcp --dport 1234 -j ACCEPT
```

### List direct rules

```
sudo firewall-cmd --direct --get-all-rules
```

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## 13. Lockdown Mode

### Enable lockdown mode

```
sudo firewall-cmd --lockdown-on
```

### Disable lockdown mode

```
sudo firewall-cmd --lockdown-off
```

### Query lockdown mode

```
sudo firewall-cmd --query-lockdown
```

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## 14. Common Real-World Examples

### Open web server ports

```
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --permanent --add-service=https
sudo firewall-cmd --reload
```

### Open MySQL only for internal network

```
sudo firewall-cmd --permanent
--add-rich-rule='rule family="ipv4" source address="192.168.1.0/24" port
port="3306" protocol="tcp" accept'
```

### Secure SSH access from one IP only

```
sudo firewall-cmd --permanent --remove-service=ssh

sudo firewall-cmd --permanent
--add-rich-rule='rule family="ipv4" source address="192.168.1.10" service
name="ssh" accept'
```

### Open custom application port

```
sudo firewall-cmd --permanent --add-port=9000/tcp
sudo firewall-cmd --reload
```

### Block attacker IP

```
sudo firewall-cmd --permanent
--add-rich-rule='rule family="ipv4" source address="203.0.113.5" drop'
```

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## 15. Troubleshooting Commands

### Check complete firewall configuration

```
sudo firewall-cmd --list-all-zones
```

### Verify service existence

```
sudo firewall-cmd --get-services
```

### Check if service is enabled

```
sudo firewall-cmd --query-service=http
```

### Check active interfaces

```
sudo firewall-cmd --get-active-zones
```

### Show runtime configuration

```
sudo firewall-cmd --runtime-to-permanent
```

### View logs

```
journalctl -u firewalld
```

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## 16. Useful Service Names

Common service names usable with `--add-service`:

| Service | Name |
|---------|------|
| SSH     | ssh  |
| HTTP    | http |

| Service        | Name           |
|----------------|----------------|
| HTTPS          | https          |
| DNS            | dns            |
| FTP            | ftp            |
| SMTP           | smtp           |
| DHCP           | dhcp           |
| Samba          | samba          |
| NFS            | nfs            |
| Cockpit        | cockpit        |
| PostgreSQL     | postgresql     |
| MySQL/MariaDB  | mysql          |
| Kubernetes API | kube-apiserver |

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## 17. Best Practices

1. Always use `--permanent` for production changes.
2. Reload firewall after permanent changes.
3. Test SSH access before removing existing SSH rules.
4. Use rich rules for granular security.
5. Limit database access to internal IPs.
6. Document custom rules.
7. Regularly review open ports and services.
8. Backup firewall configuration.

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## 18. Backup and Restore

### Backup firewalld configuration

```
sudo tar -czvf firewalld-backup.tar.gz /etc/firewalld
```

## Restore configuration

```
sudo tar -xzvf firewallld-backup.tar.gz -C /  
sudo firewall-cmd --reload
```

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# 19. Reset Firewallld

## Restore defaults

```
sudo firewall-cmd --complete-reload
```

## Remove all custom rules manually

```
sudo rm -rf /etc/firewalld/zones/*  
sudo firewall-cmd --reload
```

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# 20. Useful Workflow Example

## Example: Configure secure Linux web server

### Step 1: Allow SSH

```
sudo firewall-cmd --permanent --add-service=ssh
```

### Step 2: Allow web traffic

```
sudo firewall-cmd --permanent --add-service=http  
sudo firewall-cmd --permanent --add-service=https
```

### Step 3: Remove unnecessary services

```
sudo firewall-cmd --permanent --remove-service=cockpit
```

#### Step 4: Reload firewall

```
sudo firewall-cmd --reload
```

#### Step 5: Verify rules

```
sudo firewall-cmd --list-all
```

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## Conclusion

This guide covers:

- Basic firewall administration
- Zones
- Services
- Ports
- Rich rules
- NAT and forwarding
- Security hardening
- Troubleshooting
- Real-world administration examples

These commands are widely used in Linux system administration, server hardening, DevOps, and cybersecurity environments.